

第四节 隐函数的微分法

一、一个方程的情形

二、方程组的情形

本节讨论：

1) 方程在什么条件下才能确定隐函数．

例如, 方程 $x^2 + \sqrt{y} + C = 0$

当 $C < 0$ 时, 能确定隐函数;

当 $C > 0$ 时, 不能确定隐函数;

2) 在方程能确定隐函数时, 研究其连续性、可微性及求导方法问题．

一、一个方程的情形

定理1. 设函数 $F(x, y)$ 在点 $P(x_0, y_0)$ 的某一邻域内满足

① 具有连续的偏导数;

② $F(x_0, y_0) = 0$;

③ $F'_y(x_0, y_0) \neq 0$

则方程 $F(x, y) = 0$ 在点 x_0 的**某邻域内**可唯一确定一个单值连续函数 $y = f(x)$, 满足条件 $y_0 = f(x_0)$, 并有连续导数

$$\frac{dy}{dx} = -\frac{F'_x}{F'_y} \quad (\text{隐函数求导公式})$$

定理证明从略, 仅就求导公式推导如下:

设 $y = f(x)$ 为方程 $F(x, y) = 0$ 所确定的隐函数, 则

$$F(x, f(x)) \equiv 0$$

↓ 两边对 x 求导

$$\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} \frac{dy}{dx} \equiv 0$$

↓ 在 (x_0, y_0) 的某邻域内 $F'_y \neq 0$

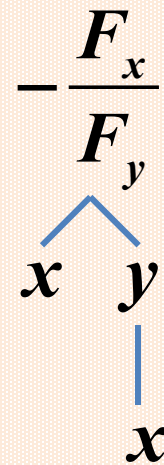
$$\frac{dy}{dx} = -\frac{F'_x}{F'_y}$$

若 $F(x, y)$ 的二阶偏导数也都连续, 则还有
二阶导数:

$$\frac{d^2 y}{dx^2} = \frac{\partial}{\partial x} \left(-\frac{F'_x}{F'_y} \right) + \frac{\partial}{\partial y} \left(-\frac{F'_x}{F'_y} \right) \frac{dy}{dx}$$

$$= -\frac{F''_{xx}F'_y - F''_{yx}F'_x}{F'^2_y} - \frac{F''_{xy}F'_y - F''_{yy}F'_x}{F'^2_y} \left(-\frac{F'_x}{F'_y} \right)$$

$$= -\frac{F''_{xx}F'^2_y - 2F''_{xy}F'_xF'_y + F''_{yy}F'^2_x}{F'^3_y}$$



$$-\frac{F'_x}{F'_y}$$

例1. 验证方程 $\sin y + e^x - xy - 1 = 0$ 在点 $(0,0)$ 某邻域可确定一个单值可导隐函数 $y = f(x)$, 并求

$$\left. \frac{dy}{dx} \right|_{x=0}, \left. \frac{d^2y}{dx^2} \right|_{x=0}$$

解: 令 $F(x, y) = \sin y + e^x - xy - 1$, 则

① $F'_x = e^x - y, F'_y = \cos y - x$ 连续,

② $F(0,0) = 0,$

③ $F'_y(0,0) = 1 \neq 0$

由定理1可知, 在 $x = 0$ 的某邻域内方程存在单值可导的隐函数 $y = f(x)$, 且

$$\left. \frac{dy}{dx} \right|_{x=0} = - \left. \frac{F'_x}{F'_y} \right|_{x=0} = - \left. \frac{e^x - y}{\cos y - x} \right|_{x=0, y=0} = -1$$

$$\left. \frac{d^2 y}{dx^2} \right|_{x=0}$$

$$= - \left. \frac{d}{dx} \left(\frac{e^x - y}{\cos y - x} \right) \right|_{x=0, y=0, y'=-1}$$

$$= - \frac{(e^x - y')(\cos y - x) - (e^x - y)(-\sin y \cdot y' - 1)}{(\cos y - x)^2} \left| \begin{array}{l} x=0 \\ y=0 \\ y'=-1 \end{array} \right.$$

$$= -3$$

导数的另一求法 — 利用隐函数求导

$$\sin y + e^x - xy - 1 = 0, \quad y = y(x)$$

两边对 x 求导

$$\cos y \cdot y' + e^x - y - xy' = 0$$

两边再对 x 求导

$$-\sin y \cdot (y')^2 + \cos y \cdot y'' + e^x - y' - y' - xy'' = 0$$

令 $x = 0$, 注意此时 $y = 0, y' = -1$

$$\left. \frac{d^2 y}{dx^2} \right|_{x=0} = -3$$

$$\begin{aligned} & y' \Big|_{x=0} \\ &= - \frac{e^x - y}{\cos y - x} \Big|_{(0,0)} \\ &= -1 \end{aligned}$$

导数的另一求法 — 微分法

方程两边同时求全微分，

$$\cos y dy + e^x dx - x dy - y dx = 0,$$

整理得，

$$dy = \frac{y - e^x}{\cos y - x} dx,$$

故，

$$f'(x) = \frac{y - e^x}{\cos y - x},$$

.....

定理2 . 若函数 $F(x, y, z)$ 满足:

- ① 在点 $P(x_0, y_0, z_0)$ 的某邻域内具有**连续偏导数**,
- ② $F(x_0, y_0, z_0) = 0$
- ③ $F'_z(x_0, y_0, z_0) \neq 0$

则方程 $F(x, y, z) = 0$ 在点 (x_0, y_0) 某一邻域内可唯一确定一个单值连续函数 $z = f(x, y)$, 满足 $z_0 = f(x_0, y_0)$, 并有连续偏导数

$$\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z}, \quad \frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z}$$

定理证明从略, 仅就求导公式推导如下:

设 $z = f(x, y)$ 是方程 $F(x, y, z) = 0$ 所确定的隐函数, 则

$$F(x, y, f(x, y)) \equiv 0$$

↓ 两边对 x 求偏导

$$F'_x + F'_z \frac{\partial z}{\partial x} \equiv 0$$

↓ 在 (x_0, y_0, z_0) 的某邻域内 $F'_z \neq 0$

$$\frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z}$$

同样可得 $\frac{\partial z}{\partial y} = -\frac{F'_y}{F'_z}$

例2. 设 $x^2 + y^2 + z^2 - 4z = 0$, 求 $\frac{\partial^2 z}{\partial x^2}$.

解法1 利用隐函数求导

$$2x + 2z \frac{\partial z}{\partial x} - 4 \frac{\partial z}{\partial x} = 0 \longrightarrow \frac{\partial z}{\partial x} = \frac{x}{2-z}$$

再对 x 求导

$$2 + 2\left(\frac{\partial z}{\partial x}\right)^2 + 2z \frac{\partial^2 z}{\partial x^2} - 4 \frac{\partial^2 z}{\partial x^2} = 0$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{1 + \left(\frac{\partial z}{\partial x}\right)^2}{2-z} = \frac{(2-z)^2 + x^2}{(2-z)^3}$$

解法2 利用公式

设 $F(x, y, z) = x^2 + y^2 + z^2 - 4z$

则 $F'_x = 2x, \quad F'_z = 2z - 4$

$$\therefore \frac{\partial z}{\partial x} = -\frac{F'_x}{F'_z} = -\frac{x}{z-2} = \frac{x}{2-z}$$

 两边对 x 求偏导

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{x}{2-z} \right) = \frac{(2-z) + x \frac{\partial z}{\partial x}}{(2-z)^2} = \frac{(2-z)^2 + x^2}{(2-z)^3}$$

解法3 微分法

方程两边同时求全微分，

$$2x\mathrm{d}x + 2y\mathrm{d}y + 2z\mathrm{d}z - 4\mathrm{d}z = 0,$$

整理得，

$$\mathrm{d}z = \frac{x}{2-z}\mathrm{d}x + \frac{y}{2-z}\mathrm{d}y,$$

故，

$$\frac{\partial z}{\partial x} = \frac{x}{2-z}, \quad \frac{\partial z}{\partial y} = \frac{y}{2-z}$$

.....

例3. 设 $F(x, y)$ 具有连续偏导数, 已知方程 $F(\frac{x}{z}, \frac{y}{z}) = 0$, 求 dz .

解法1 公式法. 设 $z = f(x, y)$ 是由方程 $F(\frac{x}{z}, \frac{y}{z}) = 0$ 确定的隐函数, 则

$$\frac{\partial z}{\partial x} = - \frac{F'_1 \cdot \frac{1}{z}}{F'_1 \cdot (-\frac{x}{z^2}) + F'_2 \cdot (-\frac{y}{z^2})} = \frac{z F'_1}{x F'_1 + y F'_2}$$

$$\frac{\partial z}{\partial y} = - \frac{F'_2 \cdot \frac{1}{z}}{F'_1 \cdot (-\frac{x}{z^2}) + F'_2 \cdot (-\frac{y}{z^2})} = \frac{z F'_2}{x F'_1 + y F'_2}$$

故 $dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = \frac{z}{x F'_1 + y F'_2} (F'_1 dx + F'_2 dy)$

解法2 微分法. 对方程两边求微分:

$$F\left(\frac{x}{z}, \frac{y}{z}\right) = 0$$

$$F_1' \cdot d\left(\frac{x}{z}\right) + F_2' \cdot d\left(\frac{y}{z}\right) = 0$$

$$F_1' \cdot \left(\frac{zdx - xdz}{z^2}\right) + F_2' \cdot \left(\frac{zdy - ydz}{z^2}\right) = 0$$

$$\frac{x F_1' + y F_2'}{z^2} dz = \frac{F_1' dx + F_2' dy}{z}$$

$$dz = \frac{z}{x F_1' + y F_2'} (F_1' dx + F_2' dy)$$

二、方程组的情形

隐函数存在定理还可以推广到方程组的情形.

以两个方程确定两个隐函数的情况为例, 即

$$\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases} \xrightarrow{\text{ }} \begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$$

由 F 、 G 的偏数组成的行列式

$$J = \frac{\partial(F, G)}{\partial(u, v)} = \begin{vmatrix} F'_u & F'_v \\ G'_u & G'_v \end{vmatrix}$$

称为 F 、 G 的雅可比(Jacobi)行列式.

定理3. 设函数 $F(x, y, u, v), G(x, y, u, v)$ 满足:

① 在点 $P(x_0, y_0, u_0, v_0)$ 的某一邻域内具有连续偏导数;

② $F(x_0, y_0, u_0, v_0) = 0, G(x_0, y_0, u_0, v_0) = 0$;

③ $J \bigg|_P = \frac{\partial(F, G)}{\partial(u, v)} \bigg|_P \neq 0$

则方程组 $F(x, y, u, v) = 0, G(x, y, u, v) = 0$ 在点 (x_0, y_0) 的某一邻域内可**唯一**确定一组满足条件 $u_0 = u(x_0, y_0), v_0 = v(x_0, y_0)$ 的**单值连续函数** $u = u(x, y), v = v(x, y)$, 且有偏导数公式:

$$\frac{\partial u}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(\underline{x}, v)} = -\frac{1}{\begin{vmatrix} F'_u & F'_v \\ G'_u & G'_v \end{vmatrix}} \begin{vmatrix} F'_x & F'_v \\ G'_x & G'_v \end{vmatrix}$$

$$\frac{\partial u}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, \underline{v})} = -\frac{1}{\begin{vmatrix} F'_u & F'_v \\ G'_u & G'_v \end{vmatrix}} \begin{vmatrix} F'_y & F'_v \\ G'_y & G'_v \end{vmatrix}$$

$$\frac{\partial v}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, \underline{x})} = -\frac{1}{\begin{vmatrix} F'_u & F'_v \\ G'_u & G'_v \end{vmatrix}} \begin{vmatrix} F'_u & F'_x \\ G'_u & G'_x \end{vmatrix}$$

$$\frac{\partial v}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, \underline{y})} = -\frac{1}{\begin{vmatrix} F'_u & F'_v \\ G'_u & G'_v \end{vmatrix}} \begin{vmatrix} F'_u & F'_y \\ G'_u & G'_y \end{vmatrix}$$

定理证明略.
仅推导偏导
数公式如下:

设方程组 $\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$ 有隐函数组 $\begin{cases} u = u(x, y) \\ v = v(x, y) \end{cases}$ 则

$$\begin{cases} F(x, y, u(x, y), v(x, y)) \equiv 0 \\ G(x, y, u(x, y), v(x, y)) \equiv 0 \end{cases}$$

两边对 x 求导得 $\begin{cases} F'_x + F'_u \cdot \frac{\partial u}{\partial x} + F'_v \cdot \frac{\partial v}{\partial x} = 0 \\ G'_x + G'_u \cdot \frac{\partial u}{\partial x} + G'_v \cdot \frac{\partial v}{\partial x} = 0 \end{cases}$

这是关于 $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}$ 的线性方程组, 在点 P 的某邻域内

系数行列式 $J = \begin{vmatrix} F'_u & F'_v \\ G'_u & G'_v \end{vmatrix} \neq 0$, 故得

$$\frac{\partial u}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(x, v)}$$

$$\frac{\partial v}{\partial x} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, x)}$$

同样可得

$$\frac{\partial u}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(y, v)}$$

$$\frac{\partial v}{\partial y} = -\frac{1}{J} \frac{\partial(F, G)}{\partial(u, y)}$$

例4. 设 $xu - yv = 0$, $yu + xv = 1$, 求 $\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}$.

解一: (隐函数求导法) 方程组两边对 x 求导, 并移项得

$$\begin{cases} x \frac{\partial u}{\partial x} - y \frac{\partial v}{\partial x} = -u \\ y \frac{\partial u}{\partial x} + x \frac{\partial v}{\partial x} = -v \end{cases}$$

由题设 $J = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2 + y^2 \neq 0$

故有 $\begin{cases} \frac{\partial u}{\partial x} = \frac{1}{J} \begin{vmatrix} -u & -y \\ -v & x \end{vmatrix} = -\frac{xu + yv}{x^2 + y^2} \\ \frac{\partial v}{\partial x} = \frac{1}{J} \begin{vmatrix} x & -u \\ y & -v \end{vmatrix} = -\frac{xv - yu}{x^2 + y^2} \end{cases}$

方程组两边对 y 求导，并移项得

$$\begin{cases} x \frac{\partial u}{\partial y} - y \frac{\partial v}{\partial y} = v \\ y \frac{\partial u}{\partial y} + x \frac{\partial v}{\partial y} = -u \end{cases}$$

由题设 $J = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2 + y^2 \neq 0$

故有 $\begin{cases} \frac{\partial u}{\partial y} = \frac{1}{J} \begin{vmatrix} v & -y \\ -u & x \end{vmatrix} = \frac{xv - yu}{x^2 + y^2} \\ \frac{\partial v}{\partial y} = \frac{1}{J} \begin{vmatrix} x & v \\ y & -u \end{vmatrix} = -\frac{xu + yv}{x^2 + y^2} \end{cases}$

解二：(微分法) 方程组两边对微分，得

$$\begin{cases} x du + u dx - y dv - v dy = 0 \\ y du + u dy + x dv + v dx = 0 \end{cases}$$

即
$$\begin{cases} x \underline{du} - y \underline{dv} = -u dx + v dy \\ y \underline{du} + x \underline{dv} = -v dx - u dy \end{cases}$$

由题设 $J = \begin{vmatrix} x & -y \\ y & x \end{vmatrix} = x^2 + y^2 \neq 0$

故有
$$\begin{cases} du = -\frac{xu + yv}{x^2 + y^2} dx + \frac{xv - yu}{x^2 + y^2} dy \\ dv = -\frac{xv - yu}{x^2 + y^2} dx - \frac{xu + yv}{x^2 + y^2} dy \end{cases}$$

练习

例1: 设 $z = f(x + y + z, xyz)$ 求 $\frac{\partial z}{\partial x}, \frac{\partial x}{\partial z}, \frac{\partial x}{\partial y}$.

$$\bullet \quad \frac{\partial z}{\partial x} = f'_1 \cdot \left(1 + \frac{\partial z}{\partial x}\right) + f'_2 \cdot (yz + xy \frac{\partial z}{\partial x})$$

$$\longrightarrow \frac{\partial z}{\partial x} = \frac{f'_1 + yzf'_2}{1 - f'_1 - xyf'_2}$$

$$\bullet \quad 1 = f'_1 \cdot \left(\frac{\partial x}{\partial z} + 1\right) + f'_2 \cdot \left(yz \frac{\partial x}{\partial z} + xy\right)$$

$$\longrightarrow \frac{\partial x}{\partial z} = \frac{1 - f'_1 - xyf'_2}{f'_1 + yzf'_2}$$

$$\bullet \quad 0 = f'_1 \cdot \left(\frac{\partial x}{\partial y} + 1\right) + f'_2 \cdot \left(yz \frac{\partial x}{\partial y} + xz\right)$$

$$\longrightarrow \frac{\partial x}{\partial y} = -\frac{f'_1 + xzf'_2}{f'_1 + yzf'_2}$$

解法2. (微分法)利用全微分形式不变性同时求出各偏导数.

$$z = f(x + y + z, xyz)$$

$$dz = f'_1 \cdot (dx + dy + dz) + f'_2 \cdot (yz dx + xz dy + xy dz)$$

解出 dx :

$$dx = \frac{-(f'_1 + xzf'_2)dy + (1 - f'_1 - xyf'_2)dz}{f'_1 + yzf'_2}$$

由 dy, dz 的系数即可得 $\frac{\partial x}{\partial y}, \frac{\partial x}{\partial z}$.

2. 设 $u = f(x, y, z)$ 有连续的一阶偏导数，
又函数 $y = y(x)$ 及 $z = z(x)$ 分别由下列两式确定：

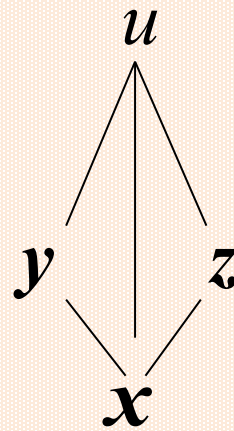
$e^{xy} - xy = 2$, $e^x = \int_0^{x-z} \frac{\sin t}{t} dt$, 求 $\frac{du}{dx}$. (考研题)

解：两个隐函数方程两边对 x 求导，得

$$\begin{cases} e^{xy}(y + xy') - (y + xy') = 0 \\ e^x = \frac{\sin(x-z)}{x-z} (1 - z') \end{cases}$$

解得 $y' = -\frac{y}{x}$, $z' = 1 - \frac{e^x(x-z)}{\sin(x-z)}$

因此 $\frac{du}{dx} = f'_1 - \frac{y}{x} f'_2 + \left[1 - \frac{e^x(x-z)}{\sin(x-z)} \right] f'_3$



3. 设 $y = y(x)$, $z = z(x)$ 是由方程 $z = x f(x + y)$ 和 $F(x, y, z) = 0$ 所确定的函数, 求 $\frac{dz}{dx}$. (考研题)

解法1 分别在各方程两端对 x 求导, 得

$$\begin{cases} z' = f + x \cdot f' \cdot (1 + y') \\ F'_x + F'_y \cdot y' + F'_z \cdot z' = 0 \end{cases} \Rightarrow \begin{cases} -x f' \cdot y' + \underline{z'} = f + x f' \\ F'_y \cdot y' + F'_z \cdot \underline{z'} = -F'_x \end{cases}$$

$$\therefore \frac{dz}{dx} = \frac{\begin{vmatrix} -x f' & f + x f' \\ F'_y & -F'_x \end{vmatrix}}{\begin{vmatrix} -x f' & 1 \\ F'_y & F'_z \end{vmatrix}} = \frac{(f + x f') F'_y - x f' \cdot F'_x}{F'_y + x f' \cdot F'_z}$$

$(F'_y + x f' \cdot F'_z \neq 0)$

解法2 微分法.

$$z = x f(x + y), \quad F(x, y, z) = 0$$

对各方程两边分别求微分:

$$\begin{cases} \mathrm{d}z = f \mathrm{d}x + x f' \cdot (\mathrm{d}x + \mathrm{d}y) \\ F_1' \mathrm{d}x + F_2' \mathrm{d}y + F_3' \mathrm{d}z = 0 \end{cases}$$

化简得

$$\begin{cases} (f + x f') \mathrm{d}x + x f' \mathrm{d}y - \mathrm{d}z = 0 \\ F_1' \mathrm{d}x + F_2' \mathrm{d}y + F_3' \mathrm{d}z = 0 \end{cases}$$

消去 $\mathrm{d}y$ **可得** $\frac{\mathrm{d}z}{\mathrm{d}x}$.

4 设 $z = f(u)$, **方程** $u = \varphi(u) + \int_y^x p(t) dt$

确定 u **是** x, y **的函数**, 其中 $f(u), \varphi(u)$ 可微, $p(t), \varphi'(u)$ 连续, 且 $\varphi'(u) \neq 1$, **求** $p(y) \frac{\partial z}{\partial x} + p(x) \frac{\partial z}{\partial y}$.

解: $\frac{\partial z}{\partial x} = f'(u) \frac{\partial u}{\partial x}, \quad \frac{\partial z}{\partial y} = f'(u) \frac{\partial u}{\partial y}$

$$\left. \begin{aligned} \frac{\partial u}{\partial x} &= \varphi'(u) \frac{\partial u}{\partial x} + p(x) \\ \frac{\partial u}{\partial y} &= \varphi'(u) \frac{\partial u}{\partial y} - p(y) \end{aligned} \right\} \Rightarrow \begin{cases} \frac{\partial u}{\partial x} = \frac{p(x)}{1 - \varphi'(u)} \\ \frac{\partial u}{\partial y} = \frac{-p(y)}{1 - \varphi'(u)} \end{cases}$$

$$\therefore p(y) \frac{\partial z}{\partial x} + p(x) \frac{\partial z}{\partial y} = f'(u) \left[p(y) \frac{\partial u}{\partial x} + p(x) \frac{\partial u}{\partial y} \right] = 0$$

内容小结

1. 隐函数(组) 存在定理

2. 隐函数 (组) 求导方法

方法1. 利用复合函数求导法则直接计算(即隐函数法) ;

方法2. 利用微分形式不变性 ;

方法3. 代公式